Master Equation and Relaxation of a Finite-State Markov Chain

Computational Physics I, PHYS 3630/5630 – Fall 2026

1. Master equation

Consider a system with a finite set of configurations (states) X. Let

$$\pi_n(X)$$

be the probability of finding the system in configuration X after the nth Markov step, and let

$$P(X \to Y)$$

be the transition probability for one step from X to Y. For every X,

$$\sum_Y P(X \to Y) = 1.$$

The probability at the next step is obtained by summing over all possible configurations from which the system can arrive at Y:

$$\boxed{\pi_{n+1}(Y) = \sum_X P(X \to Y)\, \pi_n(X).} \tag{1}$$

This is the discrete-time master equation for the Markov chain.

2. Transition matrix and the right-vector convention

Collect the probabilities into a column vector

$$\boldsymbol{\pi}_n = \begin{pmatrix} \pi_n(X_1) \\ \pi_n(X_2) \\ \vdots \\ \pi_n(X_M) \end{pmatrix}.$$

We define the transition matrix $\mathbf{P}$ by

$$\boxed{\mathbf{P}_{YX} = P(X \to Y).} \tag{2}$$

Explicitly, for M states,

$$\mathbf{P} = \begin{pmatrix} P(X_1 \to X_1) & P(X_2 \to X_1) & \cdots & P(X_M \to X_1) \\ P(X_1 \to X_2) & P(X_2 \to X_2) & \cdots & P(X_M \to X_2) \\ \vdots & \vdots & \ddots & \vdots \\ P(X_1 \to X_M) & P(X_2 \to X_M) & \cdots & P(X_M \to X_M) \end{pmatrix}.$$

Thus the *column* labels the initial state X, while the *row* labels the final state Y. In particular, the column labeled by X contains all probabilities for transitions out of X. Probability conservation gives

$$\sum_Y \mathbf{P}_{YX} = 1,$$

so $\mathbf{P}$ is column-stochastic. With this convention, probability distributions are *right vectors*, and Eq. (1) becomes

$$\boxed{\boldsymbol{\pi}_{n+1} = \mathbf{P}\boldsymbol{\pi}_n, \qquad \boldsymbol{\pi}_n = \mathbf{P}^n\boldsymbol{\pi}_0.} \tag{3}$$

3. Stationary distribution

A *stationary distribution* $\boldsymbol{\pi}_{\rm st}$ does not change after another Markov step. In physics, this is also often called the *steady-state distribution*:

$$\mathbf{P}\boldsymbol{\pi}_{\rm st} = \boldsymbol{\pi}_{\rm st}.$$

Thus the stationary distribution is a *right eigenvector* of the transition matrix with eigenvalue

$$\boxed{\lambda_1 = 1.} \tag{4}$$

For a stochastic matrix, all eigenvalues satisfy $|\lambda_a| \le 1$. We will assume that the Markov chain is irreducible and aperiodic; under the usual finite-state conditions this gives a unique stationary distribution and all other eigenvalues satisfy $|\lambda_a| < 1$.

For simplicity, suppose first that $\mathbf{P}$ can be expanded in right eigenvectors $\boldsymbol{v}_a$,

$$\mathbf{P}\boldsymbol{v}_a = \lambda_a \boldsymbol{v}_a, \qquad 1 = \lambda_1 > |\lambda_2| \ge |\lambda_3| \ge \cdots .$$

Expanding the initial distribution,

$$\boldsymbol{\pi}_0 = \boldsymbol{\pi}_{\rm st} + \sum_{a\ge 2} c_a \boldsymbol{v}_a,$$

gives

$$\boldsymbol{\pi}_n = \boldsymbol{\pi}_{\rm st} + \sum_{a\ge 2} c_a \lambda_a^n \boldsymbol{v}_a. \tag{5}$$

Every nonstationary component therefore decays as the Markov chain is iterated.

4. The second eigenvalue and relaxation time

At sufficiently long times, the slowest decaying mode is normally the one associated with the eigenvalue whose magnitude is second largest, $|\lambda_2|$. Equation (5) then has the asymptotic form

$$\boldsymbol{\pi}_n - \boldsymbol{\pi}_{\rm st} \sim \lambda_2^n \boldsymbol{v}_2.$$

Introduce a physical or Monte Carlo time

$$t = n\,\delta t.$$

For $0 < \lambda_2 < 1$,

$$\lambda_2^n = \exp(n \ln \lambda_2) = \exp\left(-\frac{t}{\tau_{\text{relax}}}\right),$$

which identifies

$$\boxed{\tau_{\text{relax}} = -\frac{\delta t}{\ln \lambda_2}.} \tag{6}$$

More generally, if the subleading eigenvalue is negative or complex, its magnitude determines the exponential envelope,

$$\boxed{\tau_{\text{relax}} = -\frac{\delta t}{\ln |\lambda_2|}.} \tag{7}$$

A negative or complex phase can additionally produce alternating or oscillatory behavior.

When λ_2 is close to one,

$$-\ln \lambda_2 \simeq 1 - \lambda_2, \qquad \tau_{\text{relax}} \simeq \frac{\delta t}{1 - \lambda_2}.$$

Thus a small spectral gap $1 - |\lambda_2|$ means slow relaxation toward the stationary distribution.

5. Relation to burn-in and autocorrelation

The relaxation time above describes how quickly the *probability distribution* $\boldsymbol{\pi}_n$ forgets its initial condition and approaches $\boldsymbol{\pi}_{\text{st}}$. It is therefore directly connected to the *burn-in* or *thermalization* period at the beginning of an MCMC simulation. Roughly,

$$N_{\text{burn}}\, \delta t \gtrsim \text{a few} \times \tau_{\text{relax}},$$

with the required factor depending on how accurately equilibrium must be reached and on the initial state.

This should be distinguished from the *autocorrelation time*. After burn-in, the chain may already be sampling the stationary distribution, but successive configurations are still correlated. For an observable $O(X)$, one studies the equilibrium autocorrelation function

$$C_O(k) = \langle \delta O_n\, \delta O_{n+k} \rangle_{\text{ss}}, \qquad \delta O_n = O(X_n) - \langle O \rangle_{\text{ss}}.$$

Its decay defines an exponential or integrated autocorrelation time. This timescale controls the statistical efficiency of measurements and motivates the number of steps n_{skip} between saved configurations (if thinning is used).

The distinction is therefore:

Timescale	**What it measures**	**Practical MCMC role**
τ_{relax}	Loss of dependence on the initial probability distribution	Burn-in / thermalization, N_{burn}
τ_{int} or τ_{corr}	Correlations between equilibrium measurements of an observable	Effective sample size; choice of n_{skip} if thinning

These two timescales are related because both ultimately arise from the eigenmodes of the same transition matrix, but they are *not the same concept*. In particular, autocorrelation is observable-dependent: an observable may have little or no overlap with the slowest eigenmode and therefore decorrelate faster than the global relaxation time suggested by $|\lambda_2|$.

6. Main picture

The Markov chain repeatedly applies the same linear operator:

$$\boldsymbol{\pi}_0 \xrightarrow{\mathbf{P}} \boldsymbol{\pi}_1 \xrightarrow{\mathbf{P}} \boldsymbol{\pi}_2 \xrightarrow{\mathbf{P}} \cdots \xrightarrow{\mathbf{P}} \boldsymbol{\pi}_{\mathrm{st}}.$$

The eigenvalue $\lambda_1 = 1$ preserves the stationary component, while all modes with $|\lambda_a| < 1$ decay. The slowest such mode, usually characterized by $|\lambda_2|$, controls the asymptotic approach to equilibrium and hence the burn-in timescale.